PRATEEK PRADEEP

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PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech, 2026, CGPA 8.51) and Python Full Stack trainee building AI-powered and full-stack applications with Python, Django REST Framework, React.js and MySQL. Built and deployed a YOLOv8 pothole detection system on a Raspberry Pi with Firebase cloud sync, presented at ICACA 2026. Seeking Software Engineer, Full-Stack or AI/ML roles.

EDUCATION

B.Tech in Computer Science | Providence College of Engineering, Chengannur, Kerala *2022 – 2026*
CGPA: 8.51 (First Class with Distinction)

Senior Secondary (CBSE) | SN Central School, Mavelikara, Kerala | GPA: 8.7 *2020 – 2022*

Secondary (CBSE) | Delhi Public School, Al Wakrah, Qatar | GPA: 8.74 *2018 – 2020*

TECHNICAL SKILLS

Languages: Python, Java, C, C#, JavaScript, Dart, Bash/Shell
Frontend: React.js, HTML5, CSS3, JavaScript, Bootstrap, Tailwind CSS
Backend: Django, Django REST Framework, REST APIs, JWT Authentication, Unit Testing (Django Test Library)
Databases: MySQL, SQLite, Oracle Database, SQL, Firebase Realtime Database
AI/ML & IoT: YOLOv8, OpenCV, Computer Vision, Model Training & Deployment, CUDA, Jupyter, Raspberry Pi Edge Deployment
Data Science: NumPy, Pandas, Matplotlib
Tools & Practices: Git, GitHub, Postman, CI/CD, Linux, Unity, OOP, Data Structures & Algorithms, Claude Code, OpenAI Codex

EXPERIENCE / TRAINING

Full Stack Development Trainee | FaithInfoTech, Technopark, Trivandrum *Ongoing*
Python Full Stack with Data Science & Generative AI Program

- Building full-stack applications with Python, Django REST Framework, React.js and MySQL/SQLite, including a Workforce Administration Portal (Users, Employees, Projects, Departments) with JWT authentication and CRUD APIs.
- Writing automated unit tests with Django Test Library; working with Git/GitHub, Postman and CI/CD workflows.

PROJECTS

Real-Time AI-Based Pothole Detection, Alert & Monitoring System *ICACA 2026*
YOLOv8, OpenCV, Python, Raspberry Pi 4, Firebase Realtime Database, REST APIs, JavaScript, Google Maps

- Trained and optimized a YOLOv8 object detection model (Jupyter, CUDA, data augmentation) to identify road hazards.
- Deployed the model on a Raspberry Pi 4 edge device, applying CLAHE and green-channel preprocessing to live video while capturing GPS data.
- Built a cloud sync pipeline over REST APIs, sending base64 event images, confidence scores and JSON metadata to Firebase.
- Developed a responsive web dashboard with live galleries and Google Maps integration to pinpoint hazard locations.

Clinic Management System (CarePoint) – Doctor Module *Team Project*
Django REST Framework, React.js, MySQL, JWT, Postman, Git/GitHub

- Built the Doctor Module of a multi-role clinic platform (Doctor, Receptionist, Pharmacy, Laboratory) using Django REST APIs and a React.js frontend.
- Implemented role-based workflows with JWT authentication: token-based patient queue, clinical examination, consultation history, lab test requests and read-only lab results.
- Designed a ~20-table MySQL schema (Role, User, Doctor, Patient, Appointment, Consultation, Prescription, LabReport) with a separate User identity linked to Doctor/Staff profiles.
- Tested APIs with Postman and collaborated with the team using Git/GitHub.

Voice-Controlled Accessible 2D Fighting Game *ICMTIF 2025*
Unity, C#, Windows Speech API

- Developed a 2D fighting game with dual-input (voice and keyboard) combat to support players with physical disabilities.
- Implemented an event-driven voice recognition system using C# dictionaries to map spoken phrases to in-game actions.
- Built custom 2D physics with Rigidbody2D: collision detection, dynamic hitboxes, state-machine animations and real-time damage calculation.

PUBLICATIONS & CERTIFICATIONS

- **Research paper accepted and presented**, ICACA 2026 Conference; project paper presented, ICMTIF 2025 Conference.
- Google Cybersecurity Professional Certificate (Coursera); Google Cloud Engineer Learning Path (GDSC).
- Ethical Hacking on IoT (CTDS India); DSA in Python (NPTEL); 2D Game Development with Unity (TILTEDU).
- IEEEXtreme 17.0 and 18.0: 2nd place in college.